

Vector Forces & Moments

Vector Concepts

Recall a vector being a quantity with magnitude and direction.

Vectors can be added with either the parallelogram method or tip to tail method, in any order.

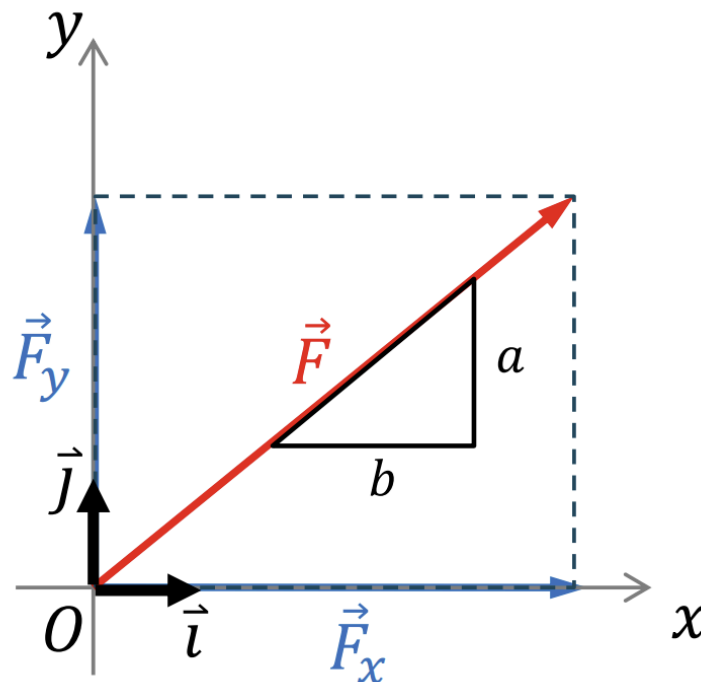
Vectors can also be decomposed into its components, in the form of

$$\vec{F} = F_x \vec{i} + F_y \vec{j}$$

where if we are given the angle θ from the horizontal,

$$F_x = F \cos(\theta), \quad F_y = F \sin(\theta)$$

Sometimes, the forces are given along a right angle triangle:



In this case, the general formula is

$$\frac{F_k}{F} = \frac{k}{\sqrt{x^2 + y^2}}$$

$$\Rightarrow F_k = \frac{k}{\sqrt{x^2 + y^2}} \times F$$

Equilibrium

Typically, if a point is stationary, it is in equilibrium, hence,

$$\sum F_x = 0$$

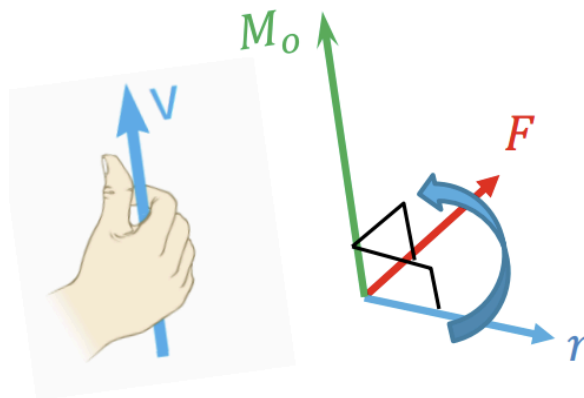
$$\sum F_y = 0$$

which allows to solve for unknown forces

Moments & Vector Multiplication

A moment is the turning effect of a force, which when taken about an axis O,

$$\vec{M}_o = \vec{r} \times \vec{F}$$



This moment can be represented as a cross product vector orthogonal to the plane containing F and r , where curling the fingers in the direction of rotation gives the direction of M .

More specifically, M is the determinant of the 3×3 matrix:

$$\vec{M}_o = \begin{vmatrix} \vec{i} & \vec{j} & \vec{k} \\ r_x & r_y & r_z \\ F_x & F_y & F_z \end{vmatrix}$$

$$= (r_y F_z - r_z F_y) \vec{i} + (r_z F_x - r_x F_z) \vec{j} + (r_x F_y - r_y F_x) \vec{k}$$

and that the magnitude is

$$|M_o| = |r||F| \sin(\theta)$$

Note that the magnitude of a vector in 3D space is

$$|\vec{a}| = \sqrt{a_x^2 + a_y^2 + a_z^2}$$

In a 2D plane, the moment vector is a magnitude multiplied to the k basis vector, or a purely orthogonal vector to the plane. Hence,

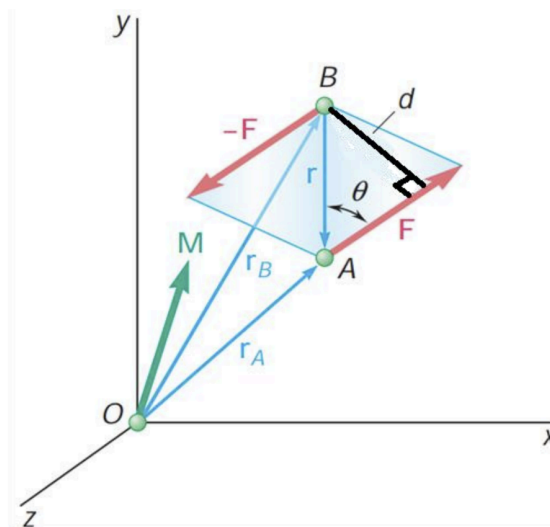
$$\vec{M}_o = (r_x F_y - r_y F_x) \vec{k}$$

the moment in a 2D plane may also be represented as a scalar, with the sign representing direction. (anticlockwise is positive)

Couples

A couple is a pair of parallel forces of equal magnitude but opposite directions, causing a pure rotation of a body.

Consider a couple in 3D space:



then, moment about O is

$$\vec{M}_o = \vec{r}_a \times \vec{F} + \vec{r}_b \times (-\vec{F}) = (\vec{r}_a - \vec{r}_b) \times \vec{F} = \vec{r} \times \vec{F}$$

hence the magnitude is

$$M_o = |r||F| \sin(\theta) = Fd$$

where d is the perpendicular distance between the two forces.

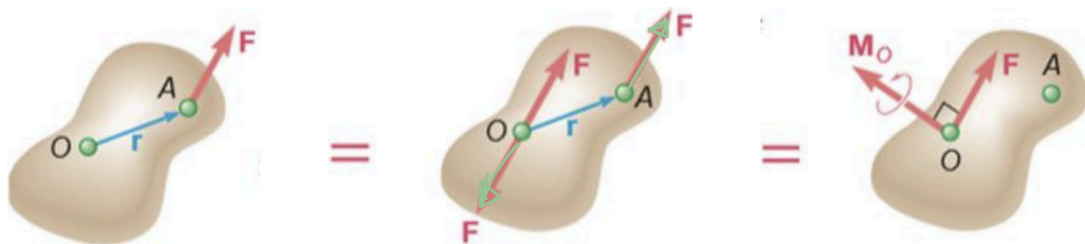
Notice how we didn't make any assumptions on where O is, hence anywhere in 3D space, a couple produces the same moment.

Force Couple Pairs

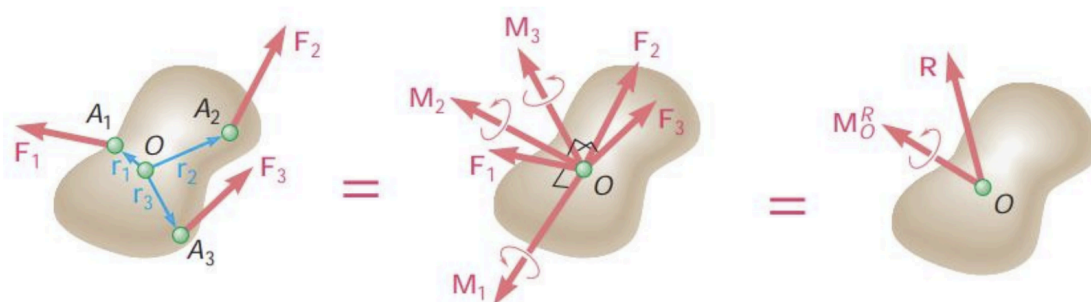
consider a force acting on a rigid body, at a certain distance away from a reference point O

We can attach two equal opposite vectors at O , as they provide no net effect, and form a couple of r equal to the distance from the force to the reference point of the rigid body.

Hence, the system can be reduced to a moment and a force F about the reference point.



The idea is that any combination of forces on a rigid body can be reduced to a force couple pair, however the moment and force will need to be equivalent

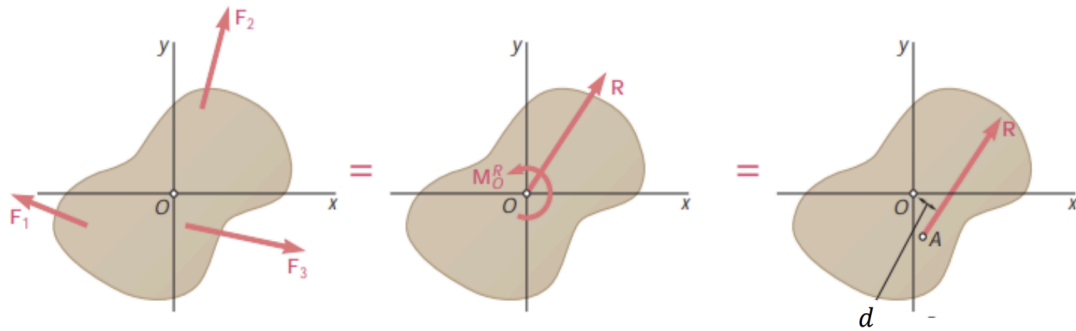


In general, for a force couple system,

$$\vec{R} = \sum \vec{F}_i$$

$$\vec{M}_O^R = \sum (\vec{r}_i \times \vec{F}_i)$$

In 2D, the resultant force R can be shifted from O to create a moment about O instead of the couple:



where the distance can be found by

$$d = \frac{M_O^R}{R}$$